

STA304 Proposal Draft — Group 12

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1. Project Title

Sleep Patterns and Their Correlates Among STA304H5 Students

2. Research Questions

- ☐ (RQ1) Is there an association between hours slept and hours spent studying per week?
- ☐ (RQ2) Does average sleep duration differ between male and female students

3. Target Population

Fall 2026 STA304H5 students, both lecture sections

4. Sampling Method

We will use stratified random sampling from our response pool. Recruitment is volunteer-based, since responses mostly come from students completing each other's surveys. We'll collect as large a pool as possible (targeting 100+) and draw a stratified random sample from within it, split by gender, keeping the male/female comparison in RQ2 balanced.

5. Data Collection Plan

The questionnaire will run on Google Forms, with the link shared in both lecture sections' student-run group chats and distributed in person before lectures (with small snacks as an incentive). The survey stays open for 10 days with a reminder at the midpoint, leaving buffer time before the checkpoint deadline to clean the data and prepare the codebook. We will monitor daily response counts; if the pool is below 100 or either gender group is below 40 by the midpoint, 2 extra collection days will be given.

6. Statistical Tests

Simple test: Two-sample t-test — used for RQ2 (male vs. female sleep duration) to test whether mean sleep hours differ by gender.

Advanced test: Multiple linear regression with model selection — used for RQ1 (association between sleep and study time) to predict sleep hours from study hours and other collected variables, using AIC-based model selection to choose the final predictors.

7. Questionnaire

8–10 questions:

1. What is your assigned course specific ID?
2. On average, how many hours do you sleep per night? (enter a number)
3. How many days per week do you have classes starting before 10:00 a.m.? (enter a number, 0–5)
4. How would you rate your overall sleep quality? Very poor / Poor / Fair / Good / Very good
5. How many hours per week do you spend studying outside of lectures and tutorials (assignments, readings, and exam preparation)? (enter a number)
6. How many days per week do you exercise for at least 30 minutes? (enter a number, 0–7)
7. How many courses are you taking this semester? (enter a number)
8. What is your current living arrangement? On-campus residence / Off-campus with family / Off-campus with roommates / Off-campus alone
9. Which lecture section are you enrolled in? LEC0101 / LEC0102
10. What is your gender? Male / Female / Other / Prefer not to disclose

Link to fill out the survey: <https://forms.cloud.microsoft/r/YAaDvr81vC?origin=lpLink>
